AN INJECTIVE MODULE WHOSE QUASI-COHERENT SHEAF HAS NONVANISHING H^1 ON A QUASI-COMPACT OPEN OF A NOETHERIAN AFFINE SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds only the topological-space remark Spec(A) = Spec(k[x,y]); all mathematics lives complete in the context span.

Editorial note. The label lemma-nonvanishing names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3789-3807: with A = k[x, y, w_1, u_1, w_2, u_2, ...]/(x^n w_n, y^n u_n, u_n^2, w_n^2) and I injective, the element 1/xy of A_{xy} is not in the image of I_x plus I_y -> I_xy (clearing denominators and multiplying by u_{n+m} w_{n+m} gives the contradiction); U = D(x) union D(y) with the Mayer-Vietoris sequence 0 -> H^0(U, M~) -> M_x plus M_y -> M_xy -> H^1(U, M~) -> 0 yields H^1(U, I~) nonzero
- Spec(A) = Spec(k[x,y]) as topological spaces (the extra variables are square-zero artefacts)

$A_{xy} \subset I_{xy}$ is not in the image of $I_x \oplus I_y \to I_{xy}$. Arguing by contradiction, suppose that

$$\frac{1}{xy} = \frac{i}{x^n} + \frac{j}{y^n}$$

for some $n \geq 1$ and $i, j \in I$. Clearing denominators we obtain

$$(xy)^{n+m-1} = x^m y^{n+m} i + x^{n+m} y^m j$$

for some $m \geq 0$. Multiplying with $u_{n+m} w_{n+m}$ we see that $u_{n+m} w_{n+m} (xy)^{n+m-1} = 0$ in A which is the desired contradiction. Let $U = D(x) \cup D(y) \subset X = \mathrm{Spec}(A)$. For any A-module M we have an exact sequence

$$0 \to H^0(U, \widetilde{M}) \to M_x \oplus M_y \to M_{xy} \to H^1(U, \widetilde{M}) \to 0$$

by Mayer-Vietoris. We conclude that $H^1(U, \widetilde{I})$ is nonzero.

Lemma 0.1. *There exists an affine scheme $X = \mathrm{Spec}(A)$ whose underlying topological space is Noetherian and an injective A-module I such that $\widetilde{I}$ has nonvanishing H^1 on some quasi-compact open U of X.*

Proof. See above. Note that $\mathrm{Spec}(A) = \mathrm{Spec}(k[x, y])$ as topological spaces. □

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